

WORK EXPERIENCE

RiVer Lab, Northeastern University

Boston, MA

Robotics Research Engineer (Prof. Taskin Padir)

May 2026 – Present

- Leading real-time pedestrian detection and tracking for an autonomous wheelchair; fused an RGB-D segmentation model with Orbbec depth data to estimate pedestrian position and velocity in unstructured outdoor environments.
- Developing a localization system for public sidewalks and indoor environments using sensor fusion and mapping techniques.

Human-Robot Interaction Lab, Northeastern University

Boston, MA

Robotics Research Engineer (Prof. Zhi Tan)

October 2025 – April 2026

- Designed and deployed a full autonomy pipeline from MuJoCo simulation to physical Stretch 3 for assistive HRI research; implemented a hierarchical macro-action system coordinating navigation, perception, and grasping as a unified mobile manipulation stack.
- Achieved reliable autonomous pick-and-place on real hardware using Nav2, LiDAR SLAM (AMCL), ArUco-based object detection, and a Jacobian IK solver; integrated speech recognition for voice-commanded task execution, demonstrated end-to-end in a 6-minute video.
- Architected a production-grade ROS2 codebase with modular components; surfaced and minimized failure modes including localization drift and contact dynamics. [\[Deployment Repo\]](#) [\[Simulation Repo\]](#)

Rivian (via RV-Tech)

Palo Alto, CA

Embedded Software Engineer (Intern)

June 2025 - August 2025

- Pioneered an automated diagnostics system for embedded vehicle software, analyzing sensor data and system logs to detect regressions during CI and release testing; eliminated manual triage and accelerated release validation.
- Deployed a root-cause analysis pipeline on AWS with LLM integration to correlate sensor recordings with connectivity logs across embedded programs and HIL systems.

Samsung Research America - Think Tank Team

Mountain View, CA

Embedded Systems Engineer

August 2023 - December 2023

- Designed a real-time embedded audio processing pipeline on a Qualcomm processor under strict latency constraints; implemented a Voice Activation Detection algorithm using FFT in C/Python and integrated it into the REX RTOS to minimize response latency.
- Prototyped a bidirectional voice pipeline enabling low-latency conversational interaction; developed and validated firmware on custom PCB hardware across full embedded stack from sensor interface to application layer.

Amazon Robotics

North Reading, MA

Robotics Software Engineer (Intern)

May 2022 - August 2022

- Developed a digital-twin of an Amazon warehouse station in Isaac Sim, simulating robots with rigid-body dynamics and photorealism; eliminated all collision risks and operational costs of training facility by transitioning operator onboarding to simulation workflows.
- Extended Isaac Sim's source code by implementing VR integration, including custom 3D transformation functions and input interfaces.
- Engineered high-fidelity robot trajectory generation using trapezoidal velocity profiles and S-curves with jerk limits.

Tufts University

Medford, MA

Research Engineer

December 2023 - May 2024

- Integrated LiDAR into embedded research platforms, developed STM32 firmware for real-time data acquisition; mentored 2 researchers

Research Engineer

2021 - December 2023

- Built and deployed an automated sensor fabrication device for Tufts NanoLab, enabling on-demand production of tension sensors.
- Accelerated prototyping cycles from weeks to hours and ensured reproducible sensor characteristics critical for research.
- Designed an interrupt-based stepper motor controller and a real-time PID controller to regulate sensor tension in embedded C.
- Presented the project at [MRS-Boston \(Tufts News\)](#); published it as first author at MDPI ([Publication Link](#))

SELECT PROJECTS

Unitree H1 Humanoid Locomotion in Isaac Lab ([Project Link](#))

- Trained a PPO locomotion policy for the Unitree H1 humanoid on rough terrain in Isaac Lab across 393M simulation steps using 4096 parallel environments; policy achieved 99.3% episode survival rate and 0.74% fall rate at terrain curriculum level 5.76/8.
- Compared bipedal vs. quadrupedal convergence dynamics against the Anymal C baseline; H1 achieved higher final reward (28.58 vs 16.30) and lower fall rate (0.74% vs 16%), attributable to a 14-term reward structure.

Quadruped Locomotion with Domain Randomization in Isaac Lab ([Project Link](#))

- Trained PPO locomotion policies for the Anymal C quadruped in Isaac Lab across 147M simulation steps using 4096 parallel environments; baseline policy converged to terrain curriculum level 6/8 with 84% episode survival rate on rough terrain.
- Conducted an ablation study on domain randomization by widening 7 physics parameters (friction 0.2–1.5, mass ± 10 kg, CoM offset, joint friction/armature, push disturbances ± 1.5 m/s) and training a second policy to quantify the robustness-performance tradeoff.
- Extended DR policy reached comparable terrain curriculum progress (5.94 vs 5.98/8) under significantly harder training conditions, confirming policy generalization over a broader physics distribution without catastrophic performance degradation.

ROS2 Multi-Sensor Perception Simulation in Isaac Sim ([Project Link](#))

- Built a multi-sensor perception simulation in Isaac Sim 5.0 featuring a 128-channel RTX LiDAR and stereo RGBD camera, streaming synchronized PointCloud2, Image, Odometry, and TF data to RViz2 via ROS2 bridge with Action Graph architecture.
- Implemented domain randomization over lighting (500–3000 lux) and LiDAR noise parameters; produced a systematic sim-to-real gap analysis documenting beam visibility artifacts, physics model limitations, and sensor rate discrepancies.

Servo Motor Feedback Controller ([Project Link](#))

- Designed and built an open-source closed-loop DC motor driver that converts DC motors into precision servos; implemented a PID controller with sub-50 μ s latency and trapezoidal trajectory planning in C on STM32, operating at 5–45 Volts and 4.1 A supply range.
- Integrated current sensing for torque feedback with high-frequency filtering and cyclic buffering; fabricated a PCB integrating H-bridge, quadrature encoder interface, and UART/I2C, validated under sensor noise and fault conditions.
- Authored a technical manuscript, submitted to *HardwareX*; preprint on SSRN ([Manuscript](#))

3D Vertebra Generation with β -Variational Autoencoder ([Project Link](#))

- Trained a 30M-parameter β -VAE on the MedShapeNet dataset to generate anatomically plausible 3D thoracic vertebrae from voxelized mesh data; achieved IoU of 0.62 on reconstruction, comparable to state-of-the-art results on analogous tasks; identified posterior collapse as the primary bottleneck via latent space diagnostics.

Incremental 3D Point Cloud Reconstruction (SfM) ([Project Link](#))

- Built an incremental Structure-from-Motion pipeline using SIFT feature detection, RANSAC-based Essential Matrix estimation, and PnP camera registration to reconstruct a colored sparse 3D point cloud from monocular images.

TECHNICAL SKILLS

- **Software & Control:** C/C++, Python, MATLAB, RTOS, ROS2, PID/Impedance Control, Git/GitLab, Inverse Kinematics
- **Robotics:** PyTorch, RL, MuJoCo/Isaac Sim/Isaac Lab, SLAM, GTSAM, Factor Graphs, OpenCV, Sensor Fusion, Kalman Filter
- **Hardware & Embedded:** STM32, ESP32, PCB Design, Sensor Integration (Encoders, Force Sensors, IMUs), I2C, SPI, UART, CAN

EDUCATION

Northeastern University

Boston, MA

Master of Science in Robotics

2024 - May 2026

- Reinforcement Learning, Feedback Control Systems, Geometric Deep Learning, Autonomous Field Robotics (3D SLAM, Kalman Filters, Sensor Fusion, GTSAM, Factor Graphs), Legged Robotics (Bipedal locomotion)

Tufts University

Medford, MA

Bachelor of Science in Electrical and Computer Engineering

2019-2023

- Related Courses: Algorithms, Computational Geometry, Data structures and C++, Embedded systems, Electronics 2, Control Systems, Probabilistic System Analysis, Parallel Computing (CUDA++)